

Clifford algebras over the field-cores of combinatorial games, and the Arf invariant as a win-bias

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June 2026

Abstract

Conway’s combinatorial games form a partially ordered abelian group under disjunctive sum but not a ring, so a Clifford algebra—which needs a commutative scalar ring—can only be built over the field-like subclasses of game-world: the surreal numbers **No**, the surcomplex numbers **No**[i], and the nimbers On_2 . We describe **pleroma**, a small library realising Clifford algebras (with nilpotents) over all three, and focus on the characteristic-2 nimber case, where the Clifford algebra is classified by the *Arf invariant* of an $\mathbb{F}_2$ quadratic form (Ovsienko). This is an experimental note. Its modest contributions are: (i) an implementation that keeps the quadratic and polar forms independent, so the char-2 theory is faithful, and that computes the Arf invariant over any nim-subfield via the field trace; (ii) the observation, verified computationally, that the Arf-bearing Gold forms $\text{Tr}(x^{1+2^a})$ are *composites of combinatorial-game operations*; and (iii) the framing of the Arf invariant as a *win-bias*: by Dickson’s democratic-value theorem it is exactly the sign of the majority margin of the form’s zero-set, which we confirm on the game-built forms. We close with a sharp open question: is there a *natural* game whose P -positions are such a form’s zero set? The classical results we lean on (the Arf–Clifford correspondence, Dickson’s theorem, the Gold-function ranks, and Turning-Corners = nim-multiplication) are reproduced by the tool as validation, not claimed as new.

1 Why three backends

A (short, partizan) combinatorial game is a pair $\{L \mid R\}$ of sets of previously-constructed games; under the disjunctive sum the games form a partially ordered abelian group [3, 2]. They do *not* form a ring: Conway’s game product is only a congruence on the *numbers*, the surreal subclass **No**. A Clifford algebra is a quotient of a tensor algebra and requires a commutative ring of scalars, so “Clifford over games” is forced onto the field-like cores of game-world:

core	structure	Clifford flavour
No (surreals)	real-closed field, char 0	the $\text{Cl}(p, q)$ theory over $\mathbb{R}$, exotic scalars
No [i] (surcomplex)	algebraically closed, char 0	the complex Clifford theory
On_2 (nimbers)	algebraically closed, char 2	genuinely different

No is real-closed [3], so finite-dimensional Clifford algebras over it are classified exactly as over $\mathbb{R}$ (the 8-fold periodicity), but one may now use infinite or infinitesimal metric entries such as $e_0^2 = \omega$, $e_1^2 = \varepsilon = \omega^{-1}$. **No**[i] is algebraically closed and gives the complex theory. The nimbers On_2 —the ordinals under nim-addition and nim-multiplication—form an algebraically closed field of *characteristic 2* [3], with the finite subfields $\mathbb{F}_{2^{2^k}}$ consisting of the nimbers below 2^{2^k} . This is the only core where Clifford theory acquires a new character, and it is our focus.

2 Characteristic 2, faithfully

In characteristic 2 a quadratic form is not determined by a symmetric bilinear form. Writing $q_i = Q(e_i)$ for the squares and $b_{ij} = Q(e_i + e_j) + Q(e_i) + Q(e_j)$ for the polar form, the relations defining the algebra are

$$e_i^2 = q_i, \quad e_i e_j + e_j e_i = b_{ij},$$

and the polar form is *alternating* ($b_{ii} = 0$) while q_i may be nonzero. If one collapses (q, b) into a single symmetric bilinear form, every char-2 algebra over an orthogonal basis becomes commutative and the content is lost. **pleroma** therefore stores q and b as independent data and reduces products by a word-rewriting routine whose only use of the sign is through the scalar's own negation; since $-1 = 1$ in On_2 , the char-2 sign-vanishing falls out automatically.

For a nonsingular quadratic form Q on $\mathbb{F}_2^{2m}$ with symplectic basis $\{a_k, b_k\}$, the *Arf invariant* is $\text{Arf}(Q) = \sum_k Q(a_k) Q(b_k) \in \mathbb{F}_2[1]$; it is independent of the basis and is a complete invariant. The crucial fact for us is the classification it controls.

Theorem (Ovsienko [8]). The classification of (real) Clifford algebras is equivalent to the classification of quadratic forms over $\mathbb{F}_2$, with the Arf invariant the complete invariant.

So computing Arf of a nim-Clifford form returns the isomorphism class of the algebra: it is the char-2 Clifford classifier, not a toy. **pleroma** computes it for a form over any nim-subfield $\mathbb{F}_{2^{2k}}$ by symplectic reduction over the field, then pushes the result to $\mathbb{F}_2$ via the trace $\text{Tr}_{\mathbb{F}_{2^{2k}}/\mathbb{F}_2}(x) = x + x^2 + \dots + x^{2^{m-1}}$, which realises the canonical isomorphism $k/\wp(k) \cong \mathbb{F}_2$. Over $\mathbb{F}_4$, for instance, the form with $q = (*2, *2)$ is of type O^- and $q = (*2, *3)$ is hyperbolic.

3 The bridge to games

Nim-addition is the disjunctive sum (Sprague–Grundy), and nim-multiplication is itself a game. Conway's *Turning Corners* has, for the single coin at (x, y) , the Grundy value given by the excludant recurrence

$$x \otimes y = \text{mex}\{(i \otimes y) \oplus (x \otimes j) \oplus (i \otimes j) : 0 \leq i < x, 0 \leq j < y\}, \quad (1)$$

which is Conway's definition of nim-multiplication [3, 2]. We implement (1) directly and verify it agrees with the algebraic Fermat-power product (computed via $F_n \otimes F_n = \frac{3}{2}F_n$ for Fermat 2-powers $F_n = 2^{2^n}$) on all pairs $x, y < 48$: two independent definitions, one combinatorial and one field-theoretic, of the same product.

This makes three operations game-realizable: the nim-product $\otimes$ (Turning Corners), the Frobenius $\square: v \mapsto v^2 = v \otimes v$ (the diagonal of Turning Corners), and $\oplus$ (disjunctive sum); hence also the trace, which is iterated $\square$ and $\oplus$.

Gold forms, and a validation. The natural quadratic forms on a char-2 field are $Q_a(x) = \text{Tr}(x \cdot x^{2^a}) = \text{Tr}(x^{1+2^a})$, the *Gold forms* [5]: $g = \text{Frob}^a$ is additive, so Q_a is genuinely quadratic. As a standard cross-check, the polar-form rank our classifier computes reproduces the known Gold-function rank $\text{rank} = m - 2 \gcd(a, m)$ (in the regime $m = 2^k$, where $m/\gcd$ is even) [7]; see Table 1.

field	a	Q	Arf	type	rank	$m - 2 \gcd(a, m)$
$\mathbb{F}_{2^4}$	1	$\text{Tr}(x^3)$	1	O^-	2	2
$\mathbb{F}_{2^8}$	1	$\text{Tr}(x^3)$	1	O^-	6	6
$\mathbb{F}_{2^8}$	2	$\text{Tr}(x^5)$	1	O^-	4	4
$\mathbb{F}_{2^{16}}$	1	$\text{Tr}(x^3)$	1	O^-	14	14
$\mathbb{F}_{2^{16}}$	4	$\text{Tr}(x^{17})$	1	O^-	8	8
$\mathbb{F}_{2^{32}}$	1	$\text{Tr}(x^3)$	1	O^-	30	30

Table 1: The classifier’s rank matches the Gold-function formula (15/15 cases up to $m = 32$; a representative selection).

The Gold forms are game-built. Because $\otimes$, $\square$, $\oplus$, and the trace are all game operations, the Gold form $Q_a(v) = \text{Tr}(v \otimes v^{2^a})$ is a composite of combinatorial-game operations applied to a position’s number value v (and under the 1-dimensional coin-turning game with single-coin Grundy value $g(n) = 2^n$, a position’s value *is* a number). We confirm this by rebuilding Q_a from literal Turning-Corners products and checking equality with the field computation on the small fields where the recurrence (1) is tractable.

Observation 1. *The quadratic forms whose Arf invariants classify the nim-Clifford algebras are not merely defined over the value field of games; they are composites of game operations. The char-2 Clifford theory is, in this constructive sense, made of games.*

4 The Arf invariant as a win-bias

The semantic payoff comes from a classical counting fact.

Theorem (Dickson [4]; Arf [1]). Over $\mathbb{F}_2$, the Arf invariant of a quadratic form is the value the form takes most often. For a nonsingular form on $\mathbb{F}_2^{2m'}$,

$$\#\{v : Q(v) = 0\} = 2^{2m'-1} + (-1)^{\text{Arf}(Q)} 2^{m'-1}.$$

Read game-theoretically: if a game had P -positions (second-player wins) exactly $\{v : Q(v) = 0\}$, then $\text{Arf}(Q)$ would be precisely the *sign of the win-bias*—which player wins from more starting positions—with a fixed margin of $2^{m'-1}$, a Gauss-sum, i.e. a square-root-scale fluctuation about 50/50. The Arf invariant is the second-order tiebreaker of an otherwise near-fair game.

We verify the count directly on the game-built Gold forms, handling the radical: for rank r and an isotropic radical, $\#\{Q = 0\} = 2^{m-1} + (-1)^{\text{Arf}} 2^{m-r/2-1}$ (and exactly 2^{m-1} when the radical is anisotropic). The brute-forced zero-counts match in every case (Table 2).

Observation 2. *On the game-built forms, the Arf invariant equals the win-bias in the counting sense: it is the sign of the majority margin of the zero-set.*

5 What remains open

The bridge closes at the level of *construction* (Observation 1) and of *counting* (the Arf invariant is the zero-set bias). What is not established is a *realisation*: a natural game whose P -positions are $\{v : Q(v) = 0\}$. The obstruction is structural and informative.

field	a	Arf	rank	$\#\{Q = 0\}$	bias
$\mathbb{F}_{2^4}$	1	1	2	4	-4
$\mathbb{F}_{2^8}$	1	1	6	112	-16
$\mathbb{F}_{2^8}$	2	1	4	96	-32
$\mathbb{F}_{2^{16}}$	1	1	14	32512	-256
$\mathbb{F}_{2^{16}}$	4	1	8	30720	-2048

Table 2: Zero-counts of the game-built Gold forms match $2^{m-1} + (-1)^{\text{Arf}} 2^{m-r/2-1}$ (bias = $\#\{Q = 0\} - 2^{m-1}$); 10/10 cases verified.

Remark 1. Normal-play disjunctive sums cannot do it. Their outcomes are $\oplus$ -linear in the components (Sprague–Grundy), so a P -position set is always a linear subspace, never a genuine quadric. This is exactly why coin-turning yields the *bilinear* nim-product but not a quadratic refinement.

Open question. *Is there a natural, necessarily non-disjunctive game whose P -positions are the zero set of a Gold form Q_a (equivalently, whose win-bias is read off by $\text{Arf}(Q_a)$)? Two candidate families avoid the linearity obstruction: interactive coin-turning / lights-out games coupled through the polar form b , and misère play, where sums are genuinely non-linear and quadratic structure can live in a misère quotient [2]. We are not aware of either in the literature; Welter’s game, the deepest relative, lands in the representation theory of symmetric groups [6] rather than in Arf invariants.*

A candidate must hit the win-bias of Table 2; `pleroma` makes that target computable. The source, tests, and the three experiments underlying Tables 1 and 2 are available at <https://github.com/a9lim/pleroma>.

Acknowledgements

The `pleroma` library, the experiments, and this note were developed in collaboration with Claude (Anthropic), which wrote the Rust/Python implementation and ran the computational probes; direction, framing, and verification are the author’s.

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